

User Story ID	Title	User Stories	Acceptance Criteria	Estimated Effort (in User Story Points)	Task ID	User Stories Task Breakdowns	Prioritization	Comments (technical and non-technical requirements, bug fixes, other)
US01	Agile Workflow and Progress Tracking	As a team, we want to manage our project using Agile / Scrum tools so that we can track progress and ensure timely delivery.	Given that the team is following Scrum, when tasks are added, updated, or completed, then the sprint board, burndown chart, and sprint documentation should reflect the current project status.	15	1	Create and continuously update sprint document for Phase 2	Must	
					2	Create and maintain fortnightly burndown charts		
					3	Update user stories, estimation efforts, acceptance criteria to reflect Phase 2 goals		
US02	Communication Practices Documentation	As a team, we want to clearly document our communication practices so that we remain aligned throughout the project.	Given that we've agreed on communication preferences, when a team member refers to the team agreement or communication plan, then they should find clearly outlined tools, channels, and protocols.	15	4	Update team agreement	Must	
					5	Update communication plan		
US03	Version Control and Collaborative Codebase Management	As a developer, I want to clearly track and record my code and be able to contribute to the codebase without conflicts, as well as see other people's work.	Given that developers are working in parallel, when code is committed and pushed to branches, then version control should prevent conflicts, and all team members should be able to review and merge changes.	15	6	Set up GitHub repository and branches	Must	
					7	Commit and updates branches to reflect work completed		
US04	Field Line Detection and Orientation Alignment	As a developer, I want to be able to detect the lines on the field, so that I have a frame of reference to determine the angle at which the field is filmed from.	Given that I upload a frame and run the offside detection module, I should see offside lines display on the same angle as the field lines.	10	9	Calculate average intersections of lines	Must	
					10	Draw line through detected points		
US05	Ball Tracking During Play	As a developer, I want to keep track of the ball when it is in play to assist in ruck and lineout offside detection.	Given that I upload pre-recorded game footage, when the ball is passed between players, a bounding box appears indicating the ball has been detected.	24	11	Ball Tracking Module	Must	
					12	Ball tracking across different angles		
					13	Determine when the ball is in possession		
US06	Player Team Classification by Jersey Colour	As a developer, I want to classify players by team based on jersey color so that we can separate them during offside detection	Given that players are visible in the frame, when the model processes jersey colors, then each player should be correctly classified as belonging to their respective team.	15	14	Detect players and extract dominant colour of jersey	Must	
					15	Compare dominant colour to detected average team colours		
					16	Sort plays based on similarity between team colours and detected dominant colours		

US07	Ruck and Lineout Detection Dataset Training	As a developer, I want to train a robust dataset to be able to detect rucks and lineouts in game footage	Given that game footage is collected and labeled, when the model is trained, then it should be able to recognize ruck and lineout formations with high accuracy across different games.	18	17	Set up Roboflow account	Must	
					18	Find or create large datasets to pre-train a model		
					19	Screenshot and clip scenes of rucks and lineouts from full game Reds footage		
US08	Kickoff Detection Dataset Training	As a developer, I want to train a robust dataset to be able to detect kickoffs in game footage	Given that labeled footage of kickoffs has been collected, when the model is trained on this dataset, then it should be able to accurately detect and label kickoff events in new game footage.	10	20	Train model on kickoff dataset	Could	
US09	Scrum Detection Dataset Training	As a developer, I want to train a robust dataset to be able to detect scrum in game footage	Given that a large, labeled dataset of scrum events is available, when the model is trained, then it should reliably identify scrums across different teams, angles, and lighting conditions in unseen footage.	10	21	Train model on scrum dataset	Could	
US10	Offside Player Movement and Tracking	As a developer, I want to keep track and visualize offside players when they are detected on the field.	Given that I upload pre-recorded game footage, when players are offside, then I should continuously see a box highlighting each player.	28	22	Player tracking modules	Must	
					23	Determine daylight zone (2m)		
					24	Field detection for offsides		
					25	Detect jersey colour of players to determine teams		
US11	Last Feet Coordinate Detection in Ruck Formations	As a developer, I want to be able to find the coordinates of last feet when players are in a ruck formation, so that I can use these coordinates as a starting point from where the offside will be monitored.	Given that I have run the player module on a ruck formation video, I should be able to track the outer most point of a player and mark it with a dot on the field.	10	26	Determine outermost point of player and mark on the field	Must	
US12	Offside Distance Measurement and Rule Validation	As a developer, I want to calculate the distance between players and last feet, so that I can measure the distance and determine offsides in rucks.	Given that I have run the player module and detected last feet, I should be able to draw a line on the field and measure the distance between the line and anyone in front/behind it.	10	27	Identify and visualize outermost point on player in ruck	Must	
					28	Visualize position of players out of ruck		
					29	Formula to measure distance between last feet and players		
US13	Web Application for User Interaction	As a developer, I want to create an intuitive web application, so that the user can easily interact with the machine learning system.	Given that I have created a web application with different pages, when information gets stored in the database, I should be able to retrieve it and populate the web application	18	30	A way to view number of offsides per game	Should	
					31	Offside penalties categorized by the type of offside that occurred		
					32	A way to view historical data from previous games	Could	

US14	Database Design for Teams and Offside Events	As a developer, I want to create a database suitable to store information about teams and offsides.	Given that I have created a database with rows to store data, when information gets processed in the ML system, it should be stored in the database.	24	33	Create database schemes	Could	Reprioritised from Must to Could based on scope change
					34	Create actual database	Won't	Reprioritised from Could to Won't based on scope change
					35	Create rows for storing offside information		
					36	Create rows for storing team data		
					37	Create rows for storing individual player data		
					38	Create rows for storing referee information		
US15	Drawing Field Markings with Accurate Perspective	As a developer, I want to detect and draw lines on the field so that we can replicate field markings accurately and at the correct angle.	Given that a frame contains visible field lines, when the line detection module runs, then it should draw lines matching actual field markings and orientation.	10	39	Implement baseline line detection of field and field markings	Should	
					40	Refine line detection with angle correction	Could	
					41	Overlay detected lines onto image	Must	
					42	Refine overlay accuracy by adjusting and testing different image conditions	Should	
US16	Entity Classification (Players, Referees, Hookers)	As a developer, I want to differentiate between people, such as players, referees and hookers, so that I can exclude certain people from analysis	Given that multiple people are in the frame, when the classification module processes the image, then it should label individuals by role and allow filtering of non-relevant participants.	8	43	Exclude player involved in lineout from lineout offside	Should	
					44	Exclude referees		
					45	Exclude people in the crowd		
US17	Post-Game Offside Event Analysis	As a developer, I want to use post-game footage to analyse potential offsides committed during that game.	Given that I have access to the Dropbox set up by my industry partner, when I open the Dropbox, then I will see post-game footage from multiple angles, which I can use to train the machine learning system on.	20	46	Counter displaying total offsides by both teams	Could	
					47	Counter displaying total offsides by Reds teams		
					48	Counter displaying total offsides by individual players		
					49	Counter displaying number of offsides given out by each referee		
US18	Automated Offside Detection for	As a head coach, I want the system to automatically detect when a player is offside	Given that I have run the pre-recorded game footage through the machine learning system, I should see	28	50	Visual indications of offsides	Must	
					51	Web application	Won't	Reprioritised from Could to Won't based on scope change
					52	Visualize ruck offsides	Must	

	Coaches	so I can easily review the footage and provide feedback.	snapshots of when offsides have occurred throughout the game.		53	Visualize lineout offsides	Must	Reprioritised from Should to Must based on scope change
					54	Visualize scrum offsides	Could	Reprioritised from Should to Could based on scope change
					55	Visualize kickoff offsides	Could	Reprioritised from Should to Could based on scope change
US19	Missed Offsides Summary for Referee Review	As a head coach, I want a summary of offsides not called by referees, so that I can educate these referees to be more alert to these types of offsides in the future.	Given that I have run the pre-recorded game footage through the machine learning system, then I will see the name of referee outputted into a folder of web console and the types and number of offsides they have missed in the game.	8	56	Visualized statistics for each referee	Could	
US20	Historical Offside Data Insights for Performance Tracking	As a head coach, I want to review statistical data on offsides across multiple games so that I can identify patterns and track improvements.	Given that a database has been created, when the machine learning system picks up an offside, then the data will be stored in the database, retrievable at any time.	8	57	Access previously stored data	Could	
US21	Model Validation with Diverse Test Data	As a developer, I want to validate my model using test data to ensure the predictions are consistent and accurate.	Given that my model is able to detect offsides, I should be able to upload new/different footage and the system should continue to detect it with little errors or uncertainties.	18	58	Train the model on footage from multiple games	Must	
					59	Train the model on footage from different weather / light conditions	Should	
					60	Train the model on footage from different angles		
US22	Robust Handling of Edge Cases in Detection	As a developer, I want the model to handle edge cases, such as obstructed view of players or varying camera angles, without providing incorrect predictions.	Given that the model has been trained on a variety of cases, when I upload an edge case, it should detect it with little inconsistencies.	10	61	Test the model on a variety of camera angles	Should	
					62	Test the model on games from different light/weather conditions		

Total Points

332